

RHEAR — every source we used

SIH 2026 · PS 26052 · DRDO / iDEX

For each: **what it says**, **why we used it**, **how it helped solve the problem**. Grouped by the decision each one informed, not alphabetically — so you can find the source behind any claim in the deck.

Roughly 45 sources. The ten marked ★ are the ones that actually changed a decision; the rest are supporting or context.

1 · Foundations — the physics our whole design obeys

★ Kuo & Morgan — Active Noise Control: A Tutorial Review

<http://www2.coe.pku.edu.cn/tpic/2010913102917710.pdf>

Says: the standard reference for active noise control. Defines FxLMS, the secondary-path problem (the speaker and air change your anti-noise before it arrives), and the **causality condition**.

Why we used it: everything in L0 is built on this.

How it helped: the causality condition became our **146 μ s budget**, which then decided the chip, the microphone type and the whole three-layer split. This is the single most load-bearing source in the project.

★ Causality study on a feedforward ANC headset with different noise directions in free field

<https://opus.lib.uts.edu.au/bitstream/10453/118176/4/Causality%20study%20on%20a%20feedforward%20active%20noise%20control%20headset%20with%20different%20noise%20directions%20in%20free%20field.pdf>

Says: whether a headset can cancel at all depends on **which direction the noise comes from** — the causal window changes with angle.

Why we used it: it told us direction is a physical constraint, not a feature.

How it helped: produced our **causal cones** result — each cup is causal over a one-sided angular range, and letting an ear use the **opposite** cup's microphone lifts coverage **38% → 68%**. That is one of our two novelty claims.

A survey on active noise control techniques — Part I: Linear systems

<https://arxiv.org/pdf/2110.00531>

Says: systematic survey of classical ANC. **Why:** to be sure we were not reinventing something standard.

How: confirmed the classical toolbox and its limits on non-stationary noise.

Toward Optimal ANC: Establishing a Mutual Information Lower Bound

<https://arxiv.org/pdf/2505.17877>

Says: an information-theoretic limit on what any controller can achieve. **Why:** to know what "good" even means. **How:** supports our use of a **predictability** measure — if the reference cannot predict the disturbance, no algorithm helps.

2 · Impulsive noise — the gunshot problem

★ Review of Advances in Active Impulsive Noise Control (Applied Sciences, 2024)

<https://doi.org/10.3390/app14031218>

Says: ordinary LMS **diverges** on impulsive noise — one huge sample makes the filter learn something catastrophically wrong. Survey of robust fixes.

Why we used it: gunfire and artillery are the defining case in our problem statement.

How it helped: gave us the robust score function in L0. We then measured that **normalisation is itself an impulse defence** — the textbook divergence only reproduces with **unnormalised** FxLMS.

Improving robustness of filtered-x least mean p-power for α -stable impulsive noise

<https://www.sciencedirect.com/science/article/abs/pii/S0003682X1100051X>

Says: models impulsive noise as α -stable and gives a p-power update rule that survives it. **Why:** the concrete algorithm behind the review above. **How:** our `score()` function.

★ MIL-STD-1474E (2015)

<https://arl.devcom.army.mil/wp-content/uploads/sites/3/2022/09/ahaah-MIL-STD-1474E-Final-15Apr2015.pdf> — superseding MIL-STD-1474D:

https://www.denix.osd.mil/soh/denix-files/sites/21/2022/12/03_MILSTD1474D-Noise-Limits.pdf

Says: the US military noise standard **dropped its simple peak-loudness rule** in favour of a model of how the ear is actually damaged.

Why we used it: it is official evidence that impulse noise is unsolved.

How it helped: this is our strongest "the problem is real" citation — a standards body changed its rule because the old one was not protecting people.

Firearms and hearing protection

<https://hearingreview.com/hearing-loss/patient-care/evaluation/firearms-and-hearing-protection>

Says: gunfire sound levels and what protection actually achieves. **Why/How:** grounds the 160 dB figure we quote.

3 · The AI-selects-the-filter family — the seed of our L2

★ SFANC — Selective Fixed-Filter ANC based on a CNN (Signal Processing, 2021)

<https://www.sciencedirect.com/science/article/abs/pii/S0165168421003546>

Says: a CNN **picks**, per frame, the best filter from a bank of pre-trained fixed filters. Fast response, no per-sample adaptation transient.

Why we used it: it shows a network can help ANC **without being in the audio path**.

How it helped: this is the direct ancestor of our **L2** — the AI writes coefficients at 62.5 Hz and never touches sound. We measured **+9.3 dB** from exactly this mechanism.

GFANC — Generative Fixed-Filter ANC (ICASSP 2023)

<https://arxiv.org/abs/2303.05788>

Says: **generates** filters by combining sub-band pieces of one broadband filter, so far less prior data is needed. **Why/How:** shows the filter bank need not be enumerated in advance — relevant if our noise classes grow.

Hybrid SFANC-FxNLMS

<https://arxiv.org/abs/2208.08082>

Says: a 1D CNN selects the filter **and** FxNLMS keeps adapting it per sample. **Why:** this is closest to what we actually built. **How:** validates combining selection with continuous adaptation rather than choosing one.

PFANC — predictive filter selection (2026)

<https://arxiv.org/pdf/2606.08171>

Says: predicts the **next** frame's filter, removing selection lag. **Why/How:** justifies L2 running ahead at 62.5 Hz rather than reacting late.

★ D-SFANC / PD-SFANC / SF-GFANC — directional filter selection (2026)

<https://arxiv.org/abs/2601.06981> · <https://arxiv.org/abs/2604.23144> · <https://arxiv.org/abs/2607.12807>

Says: use the estimated direction of the noise to choose the filter; PD-SFANC handles a **moving** source on a 36-sector grid.

Why we used it: this is our **closest prior art** — we must be able to say how we differ.

How it helped, in two ways. First, it is the honest baseline for our direction claim: they use bearing to pick a **filter**, we use it to pick a **microphone**. Second — and more useful — **PD-SFANC reports direction accuracy above 90% only at SNR $\geq$ 20 dB**. That is a quiet room, not an armoured vehicle. We designed a confidence gate so the system falls back to a fixed filter instead of steering confidently in the wrong direction.

Selective fixed-filter ANC by frequency-response matching in headphones (Applied Acoustics, 2023)

<https://www.sciencedirect.com/science/article/abs/pii/S0003682X23003031>

Says: applies filter selection specifically to **headphones** rather than ducts or rooms. **Why/How:** confirms the approach transfers to our form factor.

Neural network-based ANC algorithms: a review

<https://www.extrica.com/article/25037>

Says: survey of where neural networks are being used in ANC. **Why/How:** used to check we had not missed a family of approaches.

4 · Putting the network in the audio path — the approach we rejected

★ Deep ANC (Zhang & Wang) — Neural Networks 2021 / Interspeech 2020

<https://www.sciencedirect.com/science/article/abs/pii/S0893608021001258> · https://www.isca-archive.org/interspeech_2020/zhang20i_interspeech.pdf

Says: treat ANC as supervised learning — a network estimates the cancelling signal directly. **More than 6 dB better than FxLMS** in untrained noise.

Why we used it: this is the **strongest argument against our architecture**, so we have to be able to answer it.

How it helped: it sharpened our claim. Deep ANC is better **if** you can run a network inside the 146 μ s budget. On a ₹409 chip you cannot — we measured the compute. So we put the intelligence where the deadline is loose instead. Knowing this paper is what makes our answer credible rather than defensive.

Deep Active Speech Cancellation with a Mamba-masking network (2026)

<https://arxiv.org/abs/2502.01185>

Says: masking on the encoded reference plus multi-band phase alignment, +7.2 dB. **Why/How:** the current state of the art in the in-path approach — same trade-off, same verdict for us.

Speech-preserving deep ANC in reverberant environments (2026)

<https://arxiv.org/abs/2604.10979>

Says: adds a **speech-retention loss** that suppresses noise while **keeping** the target voice. **Why:** exactly our failure mode. **How:** shaped our asymmetric loss, which penalises deleting speech 8× harder than leaving noise.

DecNet-LMS (2026)

<https://arxiv.org/pdf/2511.03162>

Says: a fixed-weight network inverts the secondary path while an LMS filter models the primary path online, in the time domain to preserve phase. **Why/How:** another hybrid split of duties; supports our "network outside the sample loop" instinct.

5 · The small speech-enhancement models our L1 is measured against

★ GTCRN (ICASSP 2024) — 48.2 k parameters, 33.0 MMAC/s

<https://ieeexplore.ieee.org/document/10448310/> · code: <https://github.com/Xiaobin-Rong/gtcrn>

Says: a speech enhancer needing ultra-low compute, using grouped convolutions, sub-band features and a dual-path RNN.

Why we used it: it is the efficiency reference point for the whole project.

How it helped: our L1 is scaled against it — **22,956 parameters, half its size**. It is also why we rejected pruning: there is nothing left to squeeze.

LiSenNet (ICASSP 2025) — 37 k parameters

<https://ieeexplore.ieee.org/document/10888272/>

Says: sub-band modelling plus a noise detector that **skips computation** on frames with no speech.

Why/How: the source of our conditional-computation idea and the no-op gate for high-SNR frames.

DCCRN (Interspeech 2020)

<https://arxiv.org/pdf/2008.00264>

Says: heavy suppression without repairing **phase** produces metallic speech and destroys stop consonants. **Why:** fatal for call signs and coordinates. **How:** it is why we built a phase branch at all — and, ironically, our own measurement later showed our **band-level** phase rotation does more harm than good. We report both.

GTFCRN — grouped iterative CRN, INT8 on a CSK6012

<https://link.springer.com/article/10.1186/s13636-026-00455-4>

Says: a low-parameter enhancer actually deployed in INT8 on a noise-reducing-headphone chip.

Why/How: proof the class of model we chose does reach real hardware.

A Lightweight Hybrid Dual-Channel Speech Enhancement System under Low-SNR (Interspeech 2025)

<https://arxiv.org/abs/2505.19597>

Says: two-microphone enhancement tuned for low SNR. **Why/How:** low SNR is exactly where our STOI fails, so this is a live candidate for future work.

6 · Loss functions and training — where our current problem sits

★ Wang, Zhu & Kodrasi — Speech-presence probability as a secondary task (ICASSP 2021)

<https://arxiv.org/abs/2011.07547>

Says: train the network to also predict **where speech is**, weighting the two tasks by uncertainty.

Why we used it: free regularisation — the shared encoder learns "where is speech" instead of only "what to output".

How it helped: it is our **SPP head**, and it specifically protects weak consonants, which are the first thing lost in heavy suppression.

PercepNet+ — phase and SNR aware PercepNet (Interspeech 2022)

<https://arxiv.org/pdf/2203.02263> · Personalized PercepNet:

https://www.isca-archive.org/interspeech_2021/giri21_interspeech.pdf

Says: per-band gains plus a **pitch-driven comb filter** that removes noise **between** voice harmonics, at very low cost — PESQ-WB 2.54 at under 5.2% of a CPU core.

Why we used it: it looked like the direct fix for our PESQ deficit.

How it helped — by failing. We implemented it and measured **+0.004 PESQ**: nothing. That null result is what proved our problem is **distortion**, not leftover noise, and it redirected the entire project. A source that changed our direction by being wrong for us.

Learnable comb filter for lightweight full-band SE (2023)

<https://arxiv.org/abs/2306.00812>

Says: classical pitch trackers mis-estimate F0 and the comb filter then **degrades** speech; fix it by learning F0 end-to-end. **Why/How:** the mitigation for the one real failure mode of PercepNet, had we pursued it.

HC-APNet — harmonic compensation for low-complexity SE (2024)

<https://www.sciencedirect.com/science/article/abs/pii/S0167639324001328>

Says: independent confirmation of harmonic compensation at low cost. **Why/How:** cross-check on the family above.

7 · 2025-26 work on hallucination — why we will not use generative models

★ A Comparison of Generative and Discriminative Methods for Speech Enhancement: Robustness, Complexity, and Hallucination

<https://arxiv.org/html/2606.02913>

Says: discriminative models are efficient but "over-smoothed or distorted"; generative models sound best but **fabricate content**, and hallucination is **much worse at low SNR**. Also: DNSMOS scores stay high **even when content is hallucinated**.

Why we used it: a defence radio that invents a word is a safety failure.

How it helped: it is the citation behind our hard rule — **the output must remain a bounded filter of the input**, so fabrication is structurally impossible. It also warned us not to trust non-intrusive metrics

alone.

PASE / UniPASE — low-hallucination generative SE

<https://arxiv.org/pdf/2511.13300> · <https://arxiv.org/pdf/2604.14606>

Says: attempts to reduce hallucination **inside** generative enhancement. **Why/How:** named and **rejected on purpose** — still generative, still far too heavy for an MCU. Being able to say why is worth a slide.

Reducing Linguistic Hallucination in LM-Based Speech Enhancement

<https://arxiv.org/html/2605.08608>

Says: distillation to stop language-model enhancers inventing words. **Why/How:** same family, same rejection.

Interspeech 2025 URGENT Speech Enhancement Challenge

<https://arxiv.org/pdf/2505.23212>

Says: current community benchmark and its findings on robustness. **Why/How:** context for where the field is, and what evaluation is considered credible.

8 · Embedded deployment

★ Espressif ESP-DSP benchmarks

<https://docs.espressif.com/projects/esp-dsp/en/latest/esp32/esp-dsp-benchmarks.html>

Says: published CPU cycle counts for optimised DSP routines. `dsps_dotprod_s16` does 256 multiply-accumulates in **307 cycles**.

Why we used it: to decide whether the whole system fits on a ₹409 chip using the vendor's own measurements rather than our estimates.

How it helped: gave us **200 MMAC/s per core**, which produced the full timing budget — **259 of 400 MMAC/s, 65%, frame deadline PASS** — and the L0 filter length decision.

Accelerating RNN-based Speech Enhancement on a Multi-Core MCU with mixed FP16-INT8 post-training quantization

<https://arxiv.org/abs/2210.07692>

Says: how to actually get a recurrent enhancer onto a multi-core microcontroller. **Why/How:** our deployment template, and a warning that RNNs are the hard part — which our own profiling confirmed (**76% of training time is the GRU**).

Real-time SE at 3.35 ms end-to-end, 4.1 dB SI-SDRi, 376 MIPS on a low-power DSP

<https://arxiv.org/pdf/2409.18239>

Says: a working low-latency deployment with published numbers. **Why/How:** a sanity check that our latency and MIPS targets are realistic.

Feasibility of Time-Domain DNN Speech Enhancement on Embedded FPGA for Hearing Aids

<https://arxiv.org/pdf/2606.04221>

Says: the FPGA alternative to an MCU. **Why/How:** considered and set aside on cost and toolchain maturity for a student team.

ADAU1772 datasheet (Analog Devices)

<https://www.analog.com/media/en/technical-documentation/data-sheets/ADAU1772.pdf>

Says: 4 ADC / 2 DAC low-power codec; **group delay 38 μ s at fS = 192 kHz**; output configurable as line or headphone driver; digital supply from an on-chip regulator; absolute max AVDD/IOVDD **3.63 V**.

Why we used it: this is the deciding part of the whole BOM.

How it helped: every hardware decision traces here — no level shifters, no amplifier IC, a single 3.3 V rail, and the 4-ADC limit that fixed our microphone count at 2 reference + 2 error.

9 · Hearing, safety and the human case

Impact of noise on hearing in the military

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4455974/>

Says: prevalence and mechanism of noise-induced hearing loss in service personnel. **Why/How:** the evidence base for our impact slide.

Noise and Noise-Induced Hearing Loss in the Military — National Academies

<https://www.nationalacademies.org/read/11443/chapter/5>

Says: 21% noise-induced hearing loss and 28% tinnitus in post-deployment records; hearing damage is the **second most prevalent** service-connected disability. **Why/How:** turns "soldiers should hear better" into a measured, costed problem.

CDC / NIOSH hearing protection guidance

https://stacks.cdc.gov/view/cdc/111710/cdc_111710_DS1.pdf

Says: occupational hearing-protection standards. **Why/How:** the civilian market case — mining, aviation ground crew, heavy industry.

Restoring Spatial Hearing in Active Noise Control Headphones (DAGA 2025)

https://pub.dega-akustik.de/DAS-DAGA_2025/files/upload/paper/96.pdf

Says: cancelling noise must not cost you the ability to tell **where** a sound came from. **Why/How:** a safety constraint, not a feature — a soldier who cannot localise a shot is worse off. It is why we treat spatial cues as something to preserve.

Digital Augmented Reality Audio Headset — hear-through latency

<https://onlinelibrary.wiley.com/doi/10.1155/2012/457374>

Says: how much delay is tolerable when passing outside sound through to the ear. **Why/How:** bounds the hear-through path.

10 · Datasets

Source	What it gives	Licence	Link
LibriSpeech	Clean speech, speaker-disjoint splits	CC BY 4.0	https://www.openslr.org/12
MUSAN	Environmental and background noise	per-file CC / public domain	https://www.openslr.org/17

MAD — Military Audio Dataset	Gunshot, shelling, fighter, helicopter, footsteps	CC BY 4.0	https://pmc.ncbi.nlm.nih.gov/articles/
RIRS_NOISES	Real measured room impulse responses	Apache 2.0	https://www.openslr.org/28
sireNNet	Police, fire, ambulance sirens	usable, partly augmented	Kaggle
Drone corpus	Rotor / UAV noise	usable	Hugging Face
DNS Challenge	Clean speech backbone, noise variety	usable	https://github.com/microsoft/DNS-Challenge https://arxiv.org/pdf/2005.13981
NOISEX-92	Classic noise benchmark	reference only	http://mi.eng.cam.ac.uk/comp.speech/

Deliberately excluded, and this matters in the Q&A:

- **ESC-50** and **UrbanSound8K** — CC BY-**NC**, non-commercial only. A defence product is commercial.
- **DEMAND** — CC BY-**SA**, share-alike would propagate to our work.
- **MAD's communication class** — contains speech. Speech in the noise pool teaches the model to delete voices.

11 · Market and policy sources

Source	What it gives	Link
PIB, Ministry of Defence	India MoD budget FY 2025–26: ₹6,81,000 cr total, ₹1,80,000 cr capital, 75% of modernisation reserved for domestic sources	https://www.pib.gov.in/PressReleasePage.aspx?PRID=194545
MP-IDSA	Independent analysis of MoD budget estimates	https://idsa.in/publisher/issuebrief/ministry-of-defence
Fortune Business Insights	India defence electronics: \$6.85 B (2024) → \$11.35 B (2032)	https://www.fortunebusinessinsights.com/india-defence-electronics
Fortune Business Insights	Tactical headset market	https://www.fortunebusinessinsights.com/tactical-headset-market
Verified Market Reports	Tactical communication headsets: \$3.64 B (2026) → \$5.81 B (2033), 7% CAGR	https://www.verifiedmarketreports.com/product/tactical-communication-headsets/

Say "projected to" when quoting these. They are third-party forecasts, not measurements. Only RHEAR's own figures come from code we ran.

The ten that actually changed a decision

1. **Kuo & Morgan** → the 146 μ s causality budget → the entire architecture

2. **Causality study (UTS)** → causal cones → 38% → 68% coverage
3. **SFANC (2021)** → the AI writes coefficients, never audio → our L2
4. **PD-SFANC (2026)** → its $\text{SNR} \geq 20$ dB limitation → our confidence gate
5. **Deep ANC (2021)** → the strongest counter-argument → forced us to justify our split
6. **GTCRN (2024)** → the efficiency target → and why pruning is pointless for us
7. **Wang, Zhu & Kodrasi (2021)** → our speech-presence head
8. **PercepNet (2020)** → we implemented it, it gave +0.004 → **found the real problem**
9. **Generative vs Discriminative (2026)** → our bounded-mask, no-hallucination rule
10. **ESP-DSP benchmarks** → 200 MMAC/s/core → the timing budget and the ₹409 chip

MIL-STD-1474E deserves an honourable mention: it is not a method, but it is the best evidence that the problem is real and officially unsolved.